EASY · ALGEBRA

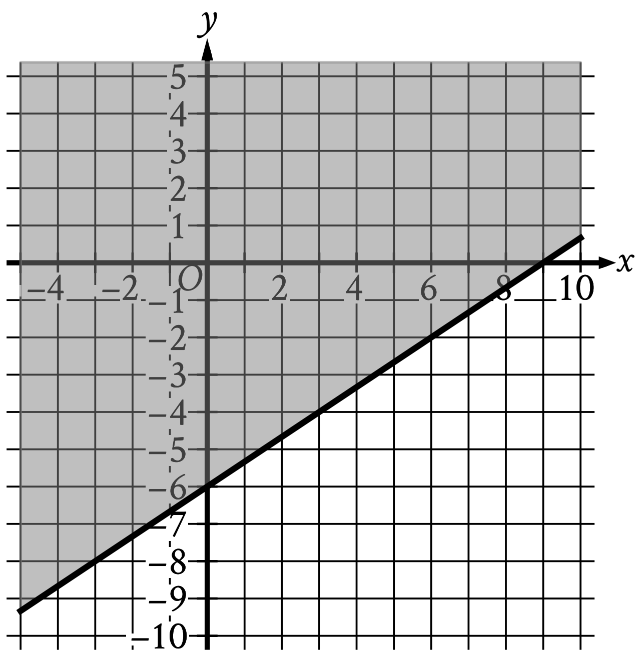
Linear inequalities in one or two variables

03

7 questions · ~11 min

SAT QUESTION BANK

2026-05-29



- The boundary of the inequality is a solid line.
    - The line slants gradually up from left to right.
    - The line passes through the following points:
        - (0 comma negative 6)
        - (9 comma 0)
- The area above and to the left of the boundary is shaded.

The shaded region shown represents the solutions to which inequality?

A. $y \geq \frac{2}{3}x - 6$

B. $y \geq \frac{2}{3}x + 6$

C. $y \geq \frac{2}{3}x - 9$

D. $y \geq \frac{2}{3}x + 9$

Q2

ALGEBRA

The point $(8, 2)$ in the xy-plane is a solution to which of the following systems of inequalities?

A. $x > 0$
$y > 0$

B. $x > 0$
$y < 0$

C. $x < 0$
$y > 0$

D. $x < 0$
$y < 0$

Q3

ALGEBRA

Tom scored 85, 78, and 98 on his first three exams in history class. Solving which inequality gives the score, G, on Tom's fourth exam that will result in a mean score on all four exams of at least 90 ?

A. $90-(85+78+98) \le 4G$

B. $4G+85+78+98 \ge 360$

C. $\frac{(G+85+78+98)}{4} \ge 90$

D. $\frac{(85+78+98)}{4} \ge 90-4G$

Q4

ALGEBRA

A geologist needs to collect at least 67 samples of lava from a volcano. If the geologist has already collected 63 samples from the volcano, what is the minimum number of additional samples the geologist needs to collect?

A. 130

B. 63

C. 4

D. 0

Q5 ALGEBRA

An elementary school teacher is ordering x workbooks and y sets of flash cards for a math class. The teacher must order at least 20 items, but the total cost of the order must not be over $80. If the workbooks cost $3 each and the flash cards cost $4 per set, which of the following systems of inequalities models this situation?

A. $$\begin{aligned} x+y &\geq 20 \\ 3x+4y &\leq 80 \end{aligned}$$

B. $$\begin{aligned} x+y &\geq 20 \\ 3x+4y &\geq 80 \end{aligned}$$

C. $$\begin{aligned} 3x+4y &\leq 20 \\ x+y &\geq 80 \end{aligned}$$

D. $$\begin{aligned} x+y &\leq 20 \\ 3x+4y &\geq 80 \end{aligned}$$

Q6 ALGEBRA

On a car trip, Rhett and Jessica each drove for part of the trip, and the total distance they drove was under 220 miles. Rhett drove at an average speed of 35 miles per hour mph, and Jessica drove at an average speed of 40 mph. Which of the following inequalities represents this situation, where r is the number of hours Rhett drove and j is the number of hours Jessica drove?

A. $35r + 40j > 220$

B. $35r + 40j < 220$

C. $40r + 35j > 220$

D. $40r + 35j < 220$

Q7 ALGEBRA

A geologist estimates that the volume of a slab of granite is greater than 12.7 cubic feet but less than 15.7 cubic feet. The geologist also estimates that the slab of granite weighs 165 pounds per cubic foot of volume. Which inequality represents this situation, where x represents the estimated total weight, in pounds, of the slab of granite?

A. $165 - 15.7 < x < 165 - 12.7$

B. $165 + 12.7 < x < 165 + 15.7$

C. $16512.7 < x < 16515.7$

D. $\frac{165}{15.7} < x < \frac{165}{12.7}$